

A Hybrid Blockchain Design Integrating Proof-of-Work and Proof-of-Quantum-Work Consensus

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Abstract—The security of today’s blockchains relies almost entirely on classical cryptography and Proof-of-Work (PoW), both of which are increasingly challenged by advances in quantum computing. Fully replacing PoW with quantum-based consensus, however, remains impractical due to hardware scarcity, high verification cost, and instability under probabilistic validation. This paper presents a hybrid blockchain design that integrates Proof-of-Work and quantum-based consensus into a unified architecture. Our system embeds quantum-validated blocks as periodic checkpoints within a predominantly classical chain, providing quantum-origin security anchors while preserving decentralization, throughput, and miner accessibility. We introduce an epoch-based consensus protocol, a hybrid chainwork formulation that balances classical and quantum contributions, and dynamic adjustment mechanisms that adapt block weights and mining difficulty as quantum hardware evolves. We further evaluate the protocol using discrete-event simulation. The results demonstrate stable protocol operation under probabilistic quantum validation, effective control of the relative contributions of classical and quantum blocks, and predictable incentive behavior across a range of operating conditions. Together, these results demonstrate that hybrid consensus offers a practical and forward-compatible path for evolving classical blockchains toward quantum-resilient operation without disruptive protocol replacement.

Index Terms—Hybrid Blockchain; Proof-of-Work; Proof-of-Quantum-Work; Quantum Computing; Consensus Protocol

I. INTRODUCTION

Blockchain systems have become a critical component of modern distributed infrastructure, supporting cryptocurrencies [1], decentralized finance [2], and large-scale digital asset management [3]. Their security and liveness today rely almost exclusively on classical cryptography and Proof-of-Work (PoW) consensus [4], where miners compete by solving hash puzzles and the longest chain determines the global state. While this design has proven robust for more than a decade, it rests on computational assumptions that are increasingly challenged by advances in quantum computing.

Quantum algorithms fundamentally alter this landscape. Shor’s algorithm [5] threatens the security of widely deployed signature schemes such as ECDSA [6], while Grover’s algorithm offers a quadratic speedup for brute-force hash inversion [7], effectively reducing PoW difficulty. Although current quantum hardware remains insufficient for practical attacks because of high noise rates and limited device scale

[8], the rapid growth of quantum devices raises an unavoidable question for blockchain architects: *how should permissionless blockchains evolve to remain secure in a future where quantum computation is accessible to adversaries?*

Traditional blockchains have begun to pursue quantum resistance by replacing vulnerable cryptographic primitives with post-quantum schemes. For example, the Quantum Resistant Ledger (QRL) replaces ECDSA with the hash-based XMSS signature standard to mitigate Shor-style attacks [9], and Algorand is integrating lattice-based FALCON signatures to protect its consensus from quantum-enabled key recovery [10]. Other efforts deploy post-quantum signing on side-chains or layer-2 networks using hash-based one-time signatures or XMSS+ [11]. However, these approaches preserve classical data structures and consensus mechanisms, often introducing liquidity fragmentation and additional trust assumptions.

An alternative is to replace PoW with a fully quantum-based Proof-of-Quantum-Work (PoQ) consensus in which block generation depends on computational tasks that are infeasible for classical hardware. For example, Boson-sampling schemes use photon detection statistics as mining puzzles verifiable via classical tests [11], and D-Wave’s PoQ prototype employs quantum annealing to generate block proofs [12]. Despite their technical promise, such systems remain limited to small-scale deployments and require scarce, high-quality quantum hardware; their high verification latency and probabilistic validation further increase fork rates, hindering deployment at scale.

These approaches represent two incompatible extremes: purely classical systems that become fragile under quantum advances, and purely quantum systems that are economically and operationally infeasible today. This paper argues that *a hybrid consensus architecture is not only a transitional compromise, but a systems necessity*. Classical PoW and quantum PoQ possess complementary properties across several dimensions: security assumptions, energy efficiency, hardware accessibility, verification cost, and mining stability. Classical PoW is easy to verify and highly decentralized, but vulnerable to future quantum advantage and energy-intensive. PoQ offers strong quantum-level security anchors and potential energy efficiency, but suffers from device scarcity, high cost, and

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slower verification. We therefore propose a *hybrid PoW-PoQ blockchain system* that integrates quantum validation into a classical blockchain without sacrificing decentralization or performance. Our design introduces periodic quantum-validated blocks that serve as cryptographic checkpoints within a predominantly classical chain. These PoQ blocks act as security anchors that cannot be forged using classical computation alone, ensuring that rewriting history beyond checkpoint boundaries requires access to quantum resources regardless of classical hash power.

The proposed design faces four main challenges in engineering such a hybrid system. First, it must ensure security under heterogeneous computation models, tolerating adversaries with either classical or quantum advantages but not both. Second, it must mitigate economic and hardware asymmetry, balancing accessibility and incentives between globally available CPUs and rare QPUs. Third, it must preserve fork stability and verification efficiency, as quantum sampling introduces latency and probabilistic confidence intervals. Lastly, it must enable long-term adaptability, evolving across decades of hardware shifts while remaining operationally stable today.

To address these challenges, we present a hybrid PoW-PoQ blockchain system that integrates quantum computation directly into consensus without excluding classical miners. The protocol operates in epochs: classical PoW blocks are mined continuously, while each epoch terminates with a quantum-validated PoQ checkpoint. These checkpoints serve as cryptographic anchors that cannot be forged without quantum hardware, bounding how far classical adversaries can rewrite history.

The system employs a hybrid chainwork model that assigns dynamically adjusted weights to PoW and PoQ blocks. A classical block contributes work proportional to its hash difficulty scaled by $\alpha(t)$, while a PoQ block contributes work scaled by $1 - \alpha(t)$, its verification fidelity, and a reward multiplier γ . As quantum hardware matures and becomes more widely accessible, the protocol gradually increases the influence of PoQ blocks, enabling a smooth transition toward quantum-dominant consensus without abrupt protocol changes.

To stabilize operation under heterogeneous conditions, the protocol also performs dynamic difficulty adjustment to maintain target block intervals and validation success rates. Both classical and quantum difficulties are periodically tuned based on observed network conditions, including quantum hardware availability, fork rates, and validation entropy. Quantum block verification uses statistical confidence scoring to tolerate noise while resisting adversarial manipulation. This design preserves classical mining participation, amortizes quantum validation overhead across epochs, and enhances security under heterogeneous adversaries.

This paper makes the following contributions:

- 1) We propose the first hybrid PoW-PoQ consensus architecture with quantum checkpoints integrated directly into classical blockchain security, without sacrificing miner inclusiveness or decentralization.

- 2) We design a chain selection mechanism that combines classical and quantum contributions using dynamic weight adjustment, reward multipliers, and fork penalties to preserve liveness and safety.
- 3) We present adaptive algorithms for tuning both classical and quantum mining difficulty, accounting for hardware heterogeneity, validation reliability, and fairness.
- 4) We show how the hybrid approach defends against future quantum supremacy attacks, reduces validation cost compared to fully quantum consensus, and ensures stable block generation under realistic conditions.
- 5) We provide a simulation-based characterization of the hybrid protocol, quantifying the frequency and persistence of quantum-validation-induced forks, demonstrating accurate regulation of block composition during both steady operation and controlled transitions, and illustrating how protocol parameters shape miner incentives.

The paper is structured as follows. Section II reviews the necessary background on classical PoW and quantum-based consensus mechanisms. Section III presents the design of the proposed hybrid PoW-PoQ consensus architecture. Section IV analyzes the security and incentive properties of the protocol under asymmetric classical and quantum adversaries. Section V discusses practical deployment considerations. Section VI evaluates the proposed design through simulation results. Section VII reviews related work, and Section VIII concludes the paper.

II. BACKGROUNDS

In this section, we introduce the necessary background of consensus mechanisms, including PoW and PoQ.

A. Traditional Consensus Mechanisms

Blockchain technology fundamentally addresses the problem: how to achieve agreement among multiple parties in the presence of malicious participants, without relying on a central trusted authority. At the core of any blockchain system lies a consensus mechanism, which is a protocol that enables a network of independent nodes to maintain a consistent, tamper-resistant ledger of transactions. The choice of consensus mechanism impacts a blockchain's security model, energy efficiency, decentralization properties, and resistance to various attack vectors [13], [14].

Traditional consensus mechanisms can be broadly categorized into several families. As the most classical consensus mechanism, Proof of Work (PoW) requires participants to expend computational resources to validate transactions [15]. Proof of Stake (PoS) selects validators based on their economic stake in the network [16]. PoW and PoS have trade-offs: PoW offers strong security and decentralization but consumes enormous energy, while PoS reduces energy consumption but risks wealth concentration.

The development of quantum computing presents both challenges and opportunities for blockchain technology. On one hand, sufficiently powerful quantum computers threaten the cryptographic foundations of existing blockchains through

Shor's algorithm [5] and could accelerate mining through Grover's algorithm [7]. On the other hand, quantum computing offers new paradigms for consensus that could address longstanding limitations of classical approaches.

B. Classical Proof of Work

PoW is introduced by Nakamoto in the Bitcoin whitepaper [15]. In PoW, miners compete to find a nonce value that, when combined with block data and processed through a cryptographic hash function, produces a hash value below a target threshold:

$$H(\text{BlockHeader}||\text{nonce}) < T, \quad (1)$$

where H is a hash function, T is the difficulty target, and $||$ denotes concatenation. The difficulty is dynamically adjusted to maintain a target block production rate (10 minutes for Bitcoin). The probability of finding a valid nonce in a single attempt is approximately $p = T/2^{N_H}$, where N_H is the hash output size in bits. This implies that miners must perform an expected $2^{N_H}/T$ hash computations per block, making the process computationally expensive but trivially verifiable.

PoW provides security through computational work. An attacker attempting to rewrite blockchain history must outpace the honest majority's computational power, i.e., the infamous "51% attack". The cumulative work in a blockchain is quantified as:

$$\text{Chainwork} = \sum_{n \in \text{chain}} W_n = \sum_n 2^{N_{\text{zeros}}(n)}, \quad (2)$$

where $N_{\text{zeros}}(n)$ represents the number of leading zero bits required in block n 's hash. The longest valid chain is considered the one that ensures all nodes on the network reach a consensus.

However, with the emerging quantum computing techniques, PoW is facing quantum vulnerability. Grover's algorithm provides a quadratic speedup for unstructured search problems [7], potentially allowing quantum computers to mine blocks $\sqrt{N}$ times faster than classical computers, given sufficient qubits. It could concentrate mining power among entities with quantum resources.

C. Quantum Hash Functions

Before we introduce the Proof of Quantum Work (PoQ), we need the concept of quantum hash functions. A quantum hash function extends classical cryptographic hashing to incorporate quantum mechanical processes [12], [17]. The general framework consists of four stages to transfer message $\mathcal{M}$ to hash value $\mathcal{H}(\mathcal{M})$:

1) *Preprocessing*: A classical hash function maps the input message $\mathcal{M}$ to quantum system parameters, $\Theta = \text{ClassicHash}(\mathcal{M})$. These parameters Θ define the Hamiltonian, evolution time, or circuit structure for the quantum computation.

2) *Unitary Evolution*: The quantum system evolves from an initial state $|\psi_0\rangle$ under a unitary operator parameterized by Θ , $|\psi_{\mathcal{M}}\rangle = U_{\Theta}|\psi_0\rangle$, resulting in a final quantum state with its corresponding pure state density matrix $\rho_{\mathcal{M}} = |\psi_{\mathcal{M}}\rangle\langle\psi_{\mathcal{M}}|$.

3) *Measurement and Witness Extraction*: Quantum measurements output statistical information about the matrix $\rho_{\mathcal{M}}$. For measurement in the computational basis, the accessible information is the probability distribution: $P_{\mathcal{M}} = \text{diagonal}(\rho_{\mathcal{M}})$. A vector $W(\rho_{\mathcal{M}})$ contains the witness operators W_{α} obtained from samples from multiple measurements.

4) *Digitalization*: Witnesses are converted to binary hash bits through thresholding:

$$H_{\alpha} = \Theta(W_{\alpha} - W_{0\alpha}) = \begin{cases} 0, & W_{\alpha} < W_{0\alpha} \\ 1, & W_{\alpha} \geq W_{0\alpha} \end{cases} \quad (3)$$

The complete hash is the concatenated bit string: $\mathcal{H}(\mathcal{M}) = H_1 H_2 \cdots H_{N_H}$.

D. Proof of Quantum Work

PoQ leverages quantum supremacy to create a consensus mechanism where mining fundamentally requires quantum computational resources. With the quantum supremacy that supports acceleration by Grover's algorithm, a performance barrier is set up between classical computing power and quantum computing power.

However, the observations of quantum states are not deterministic, caused by sampling error or hardware variations. This naturally leads to a modified chainwork definition of the block n that incorporates confidence:

$$W_n = \begin{cases} 2^{N_{\text{zeros}}(n)} - N_{\text{validator}}, & N_{\text{validator}} \leq N_{\text{max}} \\ -2^{N_{\text{zeros}}(n)}, & N_{\text{validator}} > N_{\text{max}} \end{cases}$$

where $N_{\text{validator}}$ represents the number of uncertain bits, and N_{max} is a threshold for acceptable uncertainty. This approach transforms hard validation forks (permanent chain splits) into soft forks (temporary disagreements resolved by subsequent mining), maintaining blockchain liveness despite quantum measurement uncertainty.

III. HYBRID POW-POQ CONSENSUS MECHANISM DESIGN

In this section, we present the design of the hybrid consensus mechanism.

A. Motivation for Hybrid Consensus

Pure PoW and pure PoQ represent two extremes in the design space of blockchain consensus protocols. While PoW emphasizes deployability and efficiency on classical hardware, PoQ anchors security in quantum computation but suffers from significant practical limitations. A hybrid approach enables the system to exploit their complementary strengths across security, efficiency, and deployability.

1) *Security under Quantum Advantage*: PoW becomes vulnerable under quantum advantage, as Grover's algorithm reduces the effective difficulty of hash-based mining from $O(n)$ to $O(\sqrt{n})$ [7], altering mining power distribution and long-term chain stability. PoQ, in contrast, roots its security in quantum computation itself, but its practical impact is currently constrained by the limited scale and reliability of existing quantum processors. Although today's quantum devices do not

yet pose a realistic threat to PoW-based systems, continued advances in quantum hardware are expected to gradually erode classical security margins. This motivates consensus architectures that can evolve toward quantum-resilient security without disrupting current deployments.

2) *Energy Efficiency and Hardware Availability*: Quantum computation can be orders of magnitude more energy-efficient than classical mining. However, the high capital cost of building quantum computers and maintaining large numbers of qubits limits the total available quantum computing capacity. For example, IBM currently operates fewer than 100 quantum processors with roughly 2300 qubits in total [18]. The power of the Bitcoin mining ecosystem is several orders of magnitude larger [19], implying that even with Grover’s algorithm, near-term quantum miners would have limited influence on global consensus power. These observations suggest that relying exclusively on PoQ would be economically and operationally impractical in the near term, while PoW remains inefficient from an energy perspective.

3) *Verification Latency*: PoW verification is efficient because it requires only simple hash computations that can be performed by any commodity CPU. In contrast, PoQ verification involves repeated measurements and sample aggregation to establish statistical confidence in quantum-generated proofs, resulting in higher verification latency. By confining quantum validation to periodic checkpoints, a hybrid consensus mechanism can amortize this cost over time and achieve significantly faster end-to-end verification than fully quantum-based designs.

4) *Fork Stability and Mining Efficiency*: PoW offers low fork rates and efficient verification, while PoQ introduces probabilistic validation and higher latency. The hybrid design amortizes these costs via infrequent checkpoints. Anchoring probabilistic validation to periodic quantum checkpoints allows a hybrid consensus mechanism to mitigate such divergence, improving mining efficiency while preserving quantum-based security properties.

B. PoW-PoQ Hybrid Consensus Mechanism

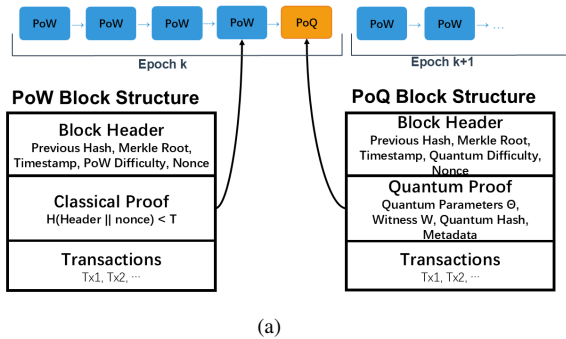


Fig. 1. An overview of the proposed hybrid blockchain structure.

We propose a hybrid PoW-PoQ consensus mechanism that combines classical PoW blocks with quantum PoQ blocks through adaptive weight adjustment.

1) *Architecture Overview*: The hybrid consensus mechanism combines classical PoW and quantum PoQ through a checkpoint-based architecture, as illustrated in Fig. 1. The protocol operates in epochs. Within each epoch, classical miners continuously generate PoW blocks, while quantum devices produce a PoQ block at the end of the epoch to serve as a quantum security anchor.

Because the aggregate computational power of classical miners is currently overwhelming, PoW blocks contribute the majority of the chainwork. PoQ blocks, in contrast, provide quantum-based security guarantees that prevent chain history within an epoch from being compromised by adversaries with quantum advantage.

Given the limited availability and computing capacity of current quantum hardware, the protocol assigns higher rewards to PoQ blocks than to PoW blocks to incentivize participation by quantum devices and ensure sufficient quantum validation capacity.

2) *Design Principles*: The system is built on three key principles.

First, it employs gradual weight adjustment. Rather than enforcing an abrupt transition from classical to quantum mining, the protocol assigns time-varying weights to different block types: PoW blocks contribute a weight of $\alpha(t)$, while PoQ blocks contribute $1 - \alpha(t)$. Initially, PoW blocks dominate chainwork, and PoQ blocks contribute only marginally. As quantum hardware matures and becomes more widely available, $\alpha(t)$ is gradually reduced to increase the influence of PoQ blocks.

Second, the protocol mandates periodic PoQ checkpoints. Each epoch must terminate with a quantum-validated PoQ block, which serves as a security anchor that cannot be forged using classical computation alone, even when classical mining dominates most blocks.

Third, the system incentivizes quantum participation through reward multipliers. Miners who generate PoQ blocks receive substantially higher rewards than those producing PoW blocks, encouraging early investment in quantum hardware while preserving low barriers to entry for classical miners.

C. Block Structure

The blockchain consists of two types of blocks. PoW blocks form the majority of the chain and are mined using traditional SHA-256 hashing [15], as in Bitcoin. Any miner with sufficient computational resources can participate in their generation. Each PoW block contains a nonce, a hash satisfying the difficulty target, and standard transaction data. These blocks are inexpensive to validate and accessible to a broad set of participants.

The protocol employs dynamically adjusted mining difficulty to ensure that a fraction $1 - \alpha(t)$ of blocks are PoQ blocks. Each PoQ block terminates an epoch, while the number of PoW blocks within an epoch is not fixed. Although epochs do not follow a strictly fixed time interval, the protocol enforces an upper bound on the number of PoW blocks that can appear between two consecutive PoQ blocks. This

guarantee ensures that, regardless of the amount of classical hash power in the system, an adversary must also possess quantum computational resources to rewrite blockchain history beyond the most recent checkpoint.

A key feature of the proposed architecture is that the proportion of PoW and PoQ blocks is adjusted dynamically over time. Specifically, the weight parameter $\alpha(t)$ evolves according to protocol-defined rules. As quantum hardware and algorithms mature, the system gradually increases the influence of PoQ blocks, enabling a smooth transition toward quantum-dominant consensus without disrupting ongoing network operation.

D. Hybrid ChainWork Design

This subsection formalizes the consensus rule used to compare competing chains in the hybrid blockchain.

We extend the classical chainwork definition in Section II-B to incorporate heterogeneous classical and quantum contributions. The hybrid blockchain defines consensus through cumulative chainwork, which aggregates heterogeneous computational contributions from classical and quantum mining. Let the blockchain consist of a sequence of blocks indexed by n . The cumulative chainwork is defined as:

$$\text{Chainwork} = \sum_{n=0}^N W_n, \quad (4)$$

where W_n denotes the work contribution of block n as follows:

$$W_n = \begin{cases} \alpha_n \cdot W_n^{\text{classical}}, & \text{PoW block} \\ (1 - \alpha_n) \cdot W_n^{\text{quantum}} \cdot \gamma, & \text{valid PoQ block} \\ -\kappa \cdot W_n^{\text{quantum}}, & \text{invalid block} \end{cases} \quad (5)$$

while $\alpha_n \in [0, 1]$ is time-dependent weight, $\gamma > 1$ is bonus multiplier for quantum validation, $\kappa \geq 1$ is penalty factor for invalid blocks.

For PoW blocks, the classical work contribution is defined as:

$$W_n^{\text{classical}} = 2^{N_{\text{zeros}}^{\text{classical}}(n)}, \quad (6)$$

where $N_{\text{zeros}}^{\text{classical}}(n)$ denotes the number of leading zero bits required by the PoW difficulty target. We distinguish classical and quantum difficulty by denoting the required number of leading zero bits as $N_{\text{zeros}}^{\text{classical}}$ and $N_{\text{zeros}}^{\text{quantum}}$, respectively. When unambiguous, we omit the superscript.

For PoQ blocks, the quantum work contribution incorporates probabilistic validation uncertainty:

$$W_n^{\text{quantum}} = \begin{cases} 2^{N_{\text{zeros}}^{\text{quantum}}(n) - N_{\text{validator}}}, & N_{\text{validator}} \leq N_{\text{max}} \\ 0, & N_{\text{validator}} > N_{\text{max}} \end{cases} \quad (7)$$

where $N_{\text{validator}}$ represents the number of uncertain bits, and N_{max} is a threshold for acceptable uncertainty.

This chainwork definition determines fork resolution and leader selection in the hybrid blockchain. We next describe how nodes execute the consensus protocol using this rule.

Algorithm 1 Hybrid PoW-PoQ Consensus Protocol (Node Operation)

```

1: Initialize local chain  $\mathcal{C}$ , epoch counter  $e \leftarrow 0$ ,  $N_{\text{max}}$ 
2: while node is active do
    Classical PoW mining, if available
3:   if number of consecutive PoW blocks  $\geq N_{\text{max}}$  then
4:     Stop PoW mining and wait for new PoQ block
5:   end if
6:   Attempt to mine PoW block under difficulty  $D_c$ 
7:   if PoW block found then
8:     Broadcast block to network
9:   end if
    Quantum PoQ mining, if available
10:  Execute quantum computation and collect samples
11:  Construct PoQ block with quantum proof
12:  if PoQ block found then
13:    Broadcast PoQ block
14:  end if
    Block reception and validation
15:  if a new block  $B$  is received then
16:    Validate PoW or PoQ proof in  $B$ 
17:    Update  $\mathcal{C}$  if needed
18:  end if
    Epoch transition
19:  if a valid PoQ checkpoint is committed to  $\mathcal{C}$  then
20:     $e \leftarrow e + 1$ 
21:    Run Adaptive Control Algorithm (Algorithm 2)
22:  end if
23: end while

```

E. Hybrid Consensus Protocol

This subsection specifies the protocol executed by each node, including block production, validation, epoch transitions, and interaction with the adaptive control mechanism. Algorithm 1 summarizes the behavior of a node participating in the hybrid PoW-PoQ consensus.

Algorithm 1 summarizes node operation under the hybrid PoW-PoQ protocol. Chain selection follows the hybrid chainwork rule defined in Section III-D. Adaptive parameter tuning is performed at epoch boundaries using Algorithm 2, described next.

F. Dynamic Adjustment

This subsection describes the adaptive control layer of the protocol, responsible for tuning consensus parameters under evolving hardware conditions.

As part of the protocol execution (Algorithm 1), the control loop is triggered at epoch boundaries to update block weights and mining difficulty using observed classical hash power, quantum participation, and validation reliability.

1) *Adaptive Mining Control Algorithm:* Algorithm 2 summarizes the adaptive control procedure executed at the end of each epoch.

The control loop jointly regulates three protocol variables: the block weight α , the classical mining difficulty D_c , and the

Algorithm 2 Adaptive Weight and Difficulty Control

Input: $H_c(n)$, $H_q(n)$, $p_f(n)$, $T_c(n)$, T_c^* , ϵ , $N_{\min}$, α_n , D_c , $N_{\max}$, η , $f(\cdot)$
Update block weight
1: $\alpha_{n+1} \leftarrow \alpha_n - \eta \cdot f\left(\frac{H_q(n)}{H_c(n)}\right)$
Adjust classical PoW difficulty
2: $D_c \leftarrow D_c \cdot \frac{T_c(n)}{T_c^*}$
Adjust quantum validation threshold
3: **if** $p_f(n) > \epsilon$ **then**
4: $N_{\max} \leftarrow N_{\max} + 1$ $\triangleright$ reduce rejection probability
5: **else if** $p_f(n) < \epsilon/2$ **then**
6: $N_{\max} \leftarrow \max(N_{\max} - 1, N_{\min})$
7: **end if**
Output: α_{n+1} , D_c , $N_{\max}$

quantum validation threshold $N_{\max}$. We next describe how each component contributes to system stability.

2) *Weight Adjustment*: The PoW weight α_n is updated as a function of quantum participation and validation reliability. Let $H_q(n)$ and $H_c(n)$ denote the observed quantum and classical mining capacities over a sliding window ending at block n . The protocol enforces:

$$\alpha_{n+1} = \alpha_n - \eta \cdot f\left(\frac{H_q(n)}{H_c(n)}\right), \quad (8)$$

where $\eta > 0$ is a smoothing parameter and $f(\cdot)$ is a bounded, monotone increasing function. As H_q/H_c increases, α decreases smoothly, shifting weight from PoW to PoQ.

3) *Difficulty Adjustment*: Classical and quantum difficulties are adjusted independently. For PoW blocks, difficulty adjustment follows the classical method to maintain a target block interval T_c .

For PoQ blocks, the quantum difficulty target is adjusted to maintain more variables. It provides a target epoch length, an acceptable validation confidence distribution, and it also bounds fork probability under noisy measurements.

Formally, let ϵ denote the acceptable PoQ validation failure rate, and p_f denote the quantum validation failure probability. Then we have

$$p_f = \Pr(N_{\text{validator}} > N_{\max}) \leq \epsilon, \quad (9)$$

for a fixed security parameter ϵ , adjusting quantum difficulty accordingly. When empirical estimates of p_f exceeds ϵ , the protocol increases $N_{\max}$ to lower the failure probability. When the mining speed is faster than expected, the protocol decreases $N_{\max}$ to increase the failure probability. This guarantees that PoQ block rejection remains rare and statistically predictable.

The control layer operates independently of block production logic and preserves protocol stability under hardware heterogeneity.

G. PoQ Block Mining

A PoQ block serves as a quantum security anchor in an otherwise classical-dominated chain. Without it, the system faces a critical vulnerability: an attacker with massive classical

hashpower but no quantum resources could rewrite history by mining a pure-classical alternative chain that is longer than the honest chain. By requiring quantum validation at regular intervals, the system ensures that attackers need quantum resources regardless of how much classical hashpower they control.

A quantum-capable miner gathers transactions from the mempool and begins the mining process. First, the miner uses the block header data to generate quantum computational parameters. These parameters define a specific quantum computation, such as the evolution of a spin glass system or another quantum supremacy problem. Second, the miner executes that quantum computation on their quantum processor, taking measurements and extracting statistical witnesses from the quantum state. These quantum witnesses are converted into a binary hash through thresholding. This quantum hash must meet the quantum difficulty requirement. Third, the miner packages all of this into a block with both classical and quantum proofs, along with detailed experimental data that allows validators to reproduce the quantum computation.

When other nodes receive this proposed checkpoint block, they must validate it. The quantum validation requires them to execute the same quantum computation when quantum capability is available, extract their own witnesses, compare their quantum hash to the proposed one, calculate a confidence metric indicating how likely the differences are due to quantum measurement uncertainty versus fraud, and accept the block only if confidence exceeds a threshold.

H. PoQ Block Intervals

In this design, the system maintains the expected block ratio through a stochastic process. Let N_w denote the number of consecutive PoW blocks between two adjacent checkpoints. To maintain the target ratio α for PoW blocks, $E[N_w] = \alpha/(1 - \alpha)$.

To prevent excessive deviation from the expected ratio and ensure security guarantees, we impose an upper bound $N_{\max}$ on consecutive PoW blocks. To ensure the realized PoW block ratio deviates from target α by at most tolerance τ ,

$$N_{\max} = \left\lceil \frac{\alpha(1 + \tau)}{(1 - \alpha)\tau} \right\rceil. \quad (10)$$

This guarantees that over any sufficiently long window, the empirical ratio of PoW blocks remains within $[\alpha - \tau, \alpha + \tau]$.

The upper bound $N_{\max}$ ensures that within any sequence of $(N_{\max} + 1)$ blocks, at least one PoQ block must appear, providing a deterministic security guarantee despite the stochastic PoQ placement.

I. PoQ Fork

Compared to PoQ forks, the probability of PoW forks is negligible in our setting; therefore, we focus on PoQ forks in this paper. Unlike PoW forks, which mainly arise from network latency and block propagation delays, PoQ forks stem from validation disagreement among honest validators.

Specifically, a PoQ fork occurs when two honest validators evaluate the same PoQ block but obtain different realizations

of $N_{\text{validator}}$, causing one to accept the block while the other rejects it. Let p_f denote the probability that an honest validator incorrectly classifies a valid PoQ block due to measurement uncertainty. The probability of disagreement between two validators is

$$p_{\text{fork}} = p_f(1 - p_f) + (1 - p_f)p_f = 2p_f(1 - p_f) \approx 2p_f, \quad (11)$$

for small p_f . Therefore, the PoQ fork probability is directly governed by the validation confidence threshold and can be controlled through appropriate parameter selection.

PoQ forks are soft forks and do not lead to permanent chain divergence. First, PoQ blocks are generated only at epoch boundaries, which confines any validation disagreement to a single block. Second, chain selection is based on cumulative hybrid chainwork, and the weight contributed by a PoQ block is bounded relative to that of subsequent PoW blocks.

Let L denote the number of PoW blocks appended after a PoQ checkpoint. The probability that a PoQ-induced fork persists beyond depth L is then exponentially bounded by $p_{\text{fork}} \cdot e^{-\lambda_c L}$, where λ_c denotes the PoW block arrival rate.

Consequently, even when a PoQ fork occurs, continued PoW mining resolves the fork rapidly, and the chain converges with probability one. Probabilistic quantum validation therefore, introduces only transient ambiguity without compromising consensus safety or liveness. PoQ forks are bounded in frequency, localized in scope, and self-healing under honest mining, ensuring that the hybrid protocol remains stable under both classical-dominant and quantum-enhanced operating conditions.

J. Parameter Sensitivity as a Deployment Concern

The hybrid consensus introduces several tunable parameters, but is intentionally designed so that no single parameter requires precise calibration to maintain safety or liveness. The PoW weight α and PoQ reward multiplier γ regulate the relative influence and incentives of classical and quantum mining, while the update step size η controls the responsiveness of adaptation. Misconfiguration of these parameters affects adoption speed and reward distribution rather than consensus correctness, and conservative choices ensure smooth evolution even under noisy or delayed estimates of network conditions.

Parameters governing probabilistic PoQ validation include the observed uncertainty statistic $N_{\text{validator}}$, the acceptance threshold N_{max} , the penalty factor κ , and the target failure bound ϵ . Together, these parameters bound the validation failure probability and the frequency of PoQ forks. Conservative thresholds increase validation difficulty, while looser thresholds increase transient disagreement. However, neither compromises long-term convergence due to hybrid chainwork resolution. Additional implicit parameters, such as epoch length and PoQ sampling budget, primarily affect efficiency and cost rather than safety. Overall, the protocol tolerates estimation error and delayed adjustment, making it suitable for decentralized deployment where precise global tuning is infeasible.

IV. INCENTIVE AND SECURITY ANALYSIS

This section studies the security and incentive properties of the hybrid PoW–PoQ consensus protocol under asymmetric classical and quantum resources. It analyzes how PoQ checkpoints amplify security and how Grover-accelerated mining affects classical stability.

A. Threat Model and Resource Asymmetry

We consider a blockchain network consisting of a set of miners, each equipped with classical computational resources, quantum computational resources, or both. Let H_c and H_q denote the aggregate classical and quantum computational capacities controlled by honest participants, respectively. We assume that in the near and medium term, quantum resources remain scarce, i.e., $H_q \ll H_c$.

PoW blocks are generated via classical hash inversion, whereas PoQ blocks are produced through quantum validation procedures and terminate epochs. Let $\alpha_n \in (0, 1)$ denote the protocol-defined weight assigned to PoW blocks in the cumulative chainwork at block height n .

Adversaries may possess both classical and quantum capabilities. Quantum-equipped miners are assumed to have access to Grover's algorithm, which provides a quadratic speedup for unstructured search, while classical miners are restricted to polynomial-time classical algorithms. We assume a partially synchronous network model and that cryptographic hash functions remain collision-resistant against classical adversaries.

B. Grover-Accelerated Mining and Classical Instability

Grover's algorithm reduces the complexity of hash inversion from $O(n)$ to $O(\sqrt{n})$, fundamentally altering the economics of PoW. In a pure PoW blockchain, this quadratic speedup allows a minority of quantum-equipped miners to obtain a disproportionate share of mining rewards, accelerating mining centralization and increasing fork instability. Classical difficulty adjustment can preserve average block intervals, but it cannot prevent concentration of effective mining power.

The proposed hybrid consensus internalizes Grover's speedup rather than attempting to suppress it. Instead of allowing quantum miners to covertly exploit PoW puzzles, the protocol channels quantum advantage into PoQ mining, where quantum computation is explicit, verifiable, and bounded in frequency. As quantum capabilities improve, the protocol gradually reduces the PoW weight $\alpha(t)$, shifting security reliance away from Grover-vulnerable classical work toward quantum-validated checkpoints. Consequently, Grover's algorithm does not undermine the hybrid protocol. Instead, it accelerates the intended transition in which quantum computation contributes directly to consensus security rather than destabilizing classical mining.

C. Security Contribution of PoQ under Resource Asymmetry

In the classical-dominant regime, PoW blocks account for most block production within epochs. However, the presence of mandatory PoQ checkpoints introduces a qualitative change in security guarantees. Rewriting blockchain history across

epoch boundaries requires recomputing PoQ blocks, which cannot be forged using classical computation alone. This leads to a security amplification effect, where even a small amount of honest quantum participation significantly increases the attack cost. Classical hashing power alone, regardless of its scale or efficiency, is insufficient to bypass PoQ checkpoints. Therefore, scarce quantum resources provide a disproportionate defensive advantage relative to their share of total computing power.

To rewrite blockchain history across epochs, an attacker must dominate both classical block production within epochs and PoQ checkpoint generation. Grover's algorithm reduces the marginal cost of classical hash inversion, but it does not eliminate the need for quantum validation. Therefore, the attacker would need to possess more than half of the total computing power of the network, both in traditional computing and quantum computing capabilities, meaning they would need to be able to perform a 51% attack using both traditional and quantum computing power.

D. Incentive Model

For base block reward R , PoW mining cost C_c , and quantum mining cost C_q , the expected profits of PoW mining are $\Pi_c = \alpha R - C_c$, and the expected profits of PoQ mining are $\Pi_q = (1 - \alpha)\gamma R - C_q$. Quantum mining is individually rational when $\Pi_q > \Pi_c$, yielding the adoption threshold $(\gamma - \alpha\gamma - \alpha)R > C_q - C_c$.

PoQ blocks are rare in the chain, but each PoQ block yields significantly higher rewards and chainwork contribution. The multiplier γ compensates for low block frequency and high hardware costs, ensuring that early quantum participation is economically rational.

Quantum miners are insulated from classical competition because increasing classical hash power does not dilute PoQ rewards. This decoupling creates a strong incentive for early quantum adopters, who face limited competition and enjoy high marginal returns per PoQ block.

The hybrid protocol admits a stable state in which classical miners dominate intra-epoch block production, while quantum miners specialize in checkpoint generation. As quantum hardware improves, increased competition among quantum miners naturally reduces per-miner profit, while protocol-level adjustments to α increase PoQ's influence on chainwork. This transition occurs smoothly without displacing classical miners or requiring protocol forks.

V. DEPLOYMENT CONSIDERATIONS

This section discusses practical considerations for deploying the proposed PoW-PoQ hybrid consensus in real-world permissionless blockchain environments. This section focuses on incremental deployment, operational constraints, and transition feasibility under realistic assumptions about quantum hardware availability.

A. Incremental Adoption in Permissionless Networks

A primary design goal of the hybrid consensus is to enable incremental deployment without coordination assumptions. In

contrast to approaches that require global cryptographic migration or trusted setup, the protocol allows classical-only nodes to remain functional participants throughout deployment.

Nodes that do not support PoQ validation can still follow the canonical chain using cumulative hybrid chainwork, while nodes with quantum capability can immediately contribute PoQ checkpoints and receive protocol-level rewards. This asymmetric participation model avoids hard forks and synchronized upgrades, which are particularly challenging in open, permissionless networks.

Importantly, security benefits from PoQ arise even under sparse adoption: the presence of occasional PoQ checkpoints already restricts deep chain rewrites, regardless of the fraction of quantum-enabled nodes.

B. Transition from Classical-Dominant to Quantum-Enhanced Operation

The protocol is explicitly designed for a long transitional period in which quantum computing resources are scarce, noisy, and heterogeneous. During this phase, PoW continues to dominate transaction throughput and block production, while PoQ contributes primarily to epoch-level security. As quantum participation increases over time, the protocol gradually shifts consensus weight through dynamic adjustment of the PoW weight parameter α . This shift does not require changes to block format, consensus rules, or cryptographic primitives, allowing the network to evolve continuously rather than through disruptive protocol replacements. This evolutionary approach contrasts with quantum-native designs that assume near-universal quantum capability and highlights the suitability of the hybrid model for near and medium term deployment.

C. Operational Constraints and Validator Heterogeneity

Real-world deployment of a hybrid PoW-PoQ consensus must accommodate substantial heterogeneity in validator capabilities and operational conditions. Access to quantum hardware is expected to be scarce, intermittent, and unevenly distributed, often relying on shared or cloud-based resources. This asymmetric participation model avoids excluding classical nodes and eliminates the need for synchronized upgrades or trusted registries of quantum devices.

PoQ validation also introduces asymmetric cost and latency due to quantum sampling, noise mitigation, and statistical verification. To prevent these factors from degrading liveness, the protocol restricts PoQ to infrequent epoch boundaries and decouples PoQ validation from transaction throughput. Validation uncertainty is handled through probabilistic acceptance thresholds that tolerate bounded disagreement across validators. As a result, slower or noisier quantum devices do not stall block production, and transient disagreement does not lead to persistent divergence.

Finally, the protocol explicitly separates security anchoring from throughput provision. Classical miners primarily contribute to transaction ordering and block frequency, while quantum miners contribute to epoch-level security guarantees. This separation limits centralization pressure arising from

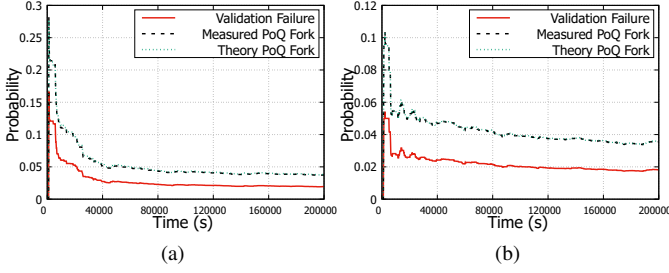


Fig. 2. Observed PoQ validation failure rate p_f and PoQ fork probability p_{fork} over time. To show the performance under different theoretical p_f , (a) shows the result when $N_{\text{max}} = 4$, and (b) shows the result when $N_{\text{max}} = 5$.

specialized hardware and prevents quantum advantage from silently dominating classical mining. By localizing the effects of probabilistic validation and bounding the influence of any single validator class, the protocol remains deployable under heterogeneous hardware, network conditions, and reliability assumptions typical of permissionless blockchain environments.

VI. SIMULATION RESULTS

A. Simulation Setting

We evaluate the proposed hybrid chainwork mechanism using a discrete-event simulator implemented in C++. The simulator explicitly models concurrent PoW and PoQ mining processes, block arrivals, validation outcomes, and adaptive chainwork weighting over time. Unless otherwise specified, all simulation results in this section are obtained using the default configuration described below.

Each simulation run spans 200000 seconds of simulated time. Events are processed using a global event queue, and statistics are aggregated over the entire run without discarding warm-up samples. Classical miners and quantum miners are modeled as independent Poisson processes. Each PoW and PoQ miner generates block attempts at a standard rate, so the effective mining capabilities H_c and H_q depend on the number of miners, which are 500 and 10, respectively. We set $\gamma = 1.5$, $\kappa = 2.0$, $N_{\text{max}} = 4$, $\alpha = 0.9$.

In the simulation section, all probabilities and rates are empirically measured from simulation runs. Since no ambiguity arises, we omit extra notations and use the same symbols as in the analytical model to denote their observed values.

B. PoQ Validation Uncertainty and Fork

We study the relationship between PoQ validation failures and PoQ fork events under probabilistic validation uncertainty. In the analytical model, a PoQ fork occurs when different honest validators reach different validity decisions on the same PoQ block. For two independent validators, the probability of such disagreement is $p_{\text{fork}} \leq 2p_f$.

In our simulator, we directly measure the observed validation failure rate as the fraction of PoQ validation events that result in rejection. We additionally measure the PoQ fork probability by tracking each PoQ block, whether at least one

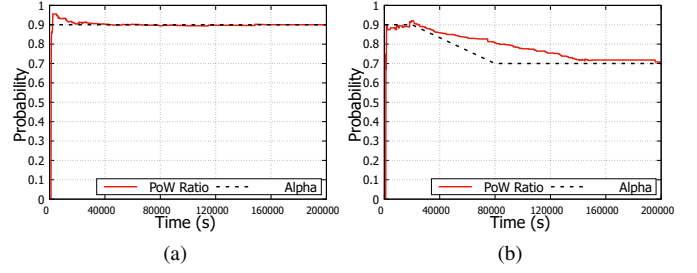


Fig. 3. Rolling PoW fraction over time. (a) shows the steady case that $\alpha = 0.9$, and (b) shows the dynamic case that α is initially 0.9, then drops to 0.7 linearly from 20000 to 80000 seconds.

validator accepts and at least one validator rejects the same block. This definition captures validator-level disagreement and corresponds directly to the analytical notion of PoQ forks induced by validation uncertainty.

Figure 2(a) plots both p_f and p_{fork} as functions of time. For comparison, the figure computes theory PoQ fork probability by directly computing $2p_f(1 - p_f)$. The results show that p_{fork} closely follows the theoretical values, validating the analytical model. Importantly, this fork probability reflects logical forks caused by disagreement on block validity rather than chain-level forks caused by competing block proposals or network latency. Figure 2(b) changes N_{max} from 4 to 5. While the validation failure probability is lower, p_{fork} converges to approximate $2p_f(1 - p_f)$, approximately $2p_f$ since p_f is low.

We additionally examined the persistence depth of PoQ forks, defined as the number of subsequent blocks required for the network to reconverge to a single chain following a PoQ fork. Across all tested configurations, we observe that PoQ forks resolve immediately or after at most one subsequent block (PoW or PoQ), yielding persistence depths of 0 or 1. This indicates that PoQ forks are short-lived and self-healing in practice, and therefore do not pose a significant threat to chain stability. These results validate the claim that PoQ forks are probabilistic, bounded, and operationally benign.

C. Steady and Dynamic Control of PoW Throughput

Next, we examine whether the protocol maintains the intended balance between PoW throughput and PoQ checkpoints. Figure 3(a) considers the steady-state behavior with a fixed control parameter $\alpha = 0.9$. The rolling fraction of PoW blocks on the canonical chain converges tightly to the target value, demonstrating that the epoch mechanism enforces the intended long-term PoW/PoQ balance despite stochastic block arrivals and probabilistic PoQ validation.

Figure 3(b) evaluates the system's response to a time-varying control parameter. Here, α is linearly decreased from 0.9 to 0.7 between 20000 and 80000 seconds. The observed PoW fraction, measured over a rolling window of 500 blocks to discount historical data, follows this change with a moderate and predictable lag, converging to approximately 0.72 by time 150000 seconds. This delay arises from epoch boundaries and the finite PoW run length per epoch, rather than from

instability in the control mechanism. These results confirm that the hybrid protocol provides both steady-state accuracy and dynamic controllability. PoW throughput neither dominates nor is overwhelmed by PoQ checkpointing, and the system smoothly adapts to changes in α while preserving stable transaction processing capacity.

D. Incentive Under Variations

Figure 4 illustrates the incentive $\mathcal{I}$ received by all PoQ miners in the hybrid chainwork mechanism under variations of the quantum bonus parameter γ , the penalty parameter κ , and the relative quantum participation ratio H_q/H_c . Figure 4(a) plots the empirical incentive $\mathcal{I}$ as a function of the bonus multiplier γ . The incentive increases monotonically and approximately linearly with γ , matching the analytical design of the hybrid chainwork model. Figure 4(b) shows the incentive $\mathcal{I}$ as a function of the penalty factor κ . As expected, increasing κ leads to a monotonic decrease in incentive, reflecting the higher cost imposed on failed or invalid PoQ attempts. The smooth decline demonstrates that κ discourages unreliable quantum participation. Figure 4(c) plots incentive versus the quantum-to-classical capacity ratio H_q/H_c . As quantum participation increases, the incentive rises steadily, despite the adaptive adjustment of the PoW weight parameter α . This result confirms that the adaptive mechanism successfully reallocates chainwork weight from PoW to PoQ as quantum capacity grows, thereby preserving incentive compatibility without destabilizing block production.

Taken together, these results validate the core incentive claims of the proposed hybrid chainwork mechanism. The parameters γ, κ control over reward amplification and risk penalization, while the dynamic adjustment driven by H_q/H_c ensures that incentives remain aligned as quantum mining capacity evolves.

VII. RELATED WORK

A. Proof-of-Work Consensus and Threats

PoW is the dominant consensus mechanism in permissionless blockchains due to its strong Sybil resistance and security guarantees [15]. However, PoW fundamentally relies on the assumed hardness of cryptographic hash inversion under classical computation. The introduction of quantum Grover's algorithm reduces the complexity of search from $O(n)$ to $O(\sqrt{n})$, effectively halving the security margin of hash-based PoW systems [7]. This observation has motivated extensive research into post-quantum cryptography and quantum-resistant blockchain primitives. Several works have examined the implications of quantum attacks on blockchain security, focusing on reduced difficulty margins, increased fork rates, and altered incentive structures [20]. These studies highlight that cryptographic hardness alone is insufficient to ensure long-term decentralization in the presence of quantum advantage. Survey works emphasize replacing vulnerable cryptographic components, such as public-key signatures and key exchange, with lattice-based, hash-based, or code-based alternatives [21]. While these approaches improve cryptographic robustness,

they generally leave the consensus mechanism unchanged and do not address the asymmetric coexistence of classical and quantum miners.

B. Post-Quantum Cryptography in Blockchain Systems

Recently, several blockchain systems have experimented with post-quantum signatures and authentication mechanisms, such as XMSS-based transaction signing, lattice-based schemes, and hybrid classical–post-quantum authentication [22]–[24]. While these approaches strengthen cryptographic primitives, recent evaluations emphasize that they leave mining and consensus unchanged and therefore do not mitigate quantum-induced mining asymmetry [21], [25]. As a result, post-quantum cryptography alone does not prevent quantum miners from gaining disproportionate influence in PoW-based systems. It motivates the design of schemes that explicitly integrate quantum computing into the consensus mechanism itself.

C. Quantum Key Distribution

Quantum Key Distribution (QKD) provides information-theoretic security for key exchange and has been explored for securing blockchain communication [26]–[28]. However, QKD addresses communication security rather than consensus security and requires dedicated quantum communication infrastructure, specialized hardware, and stable pairwise links, which limits scalability. Consequently, despite extensive research, QKD has seen minimal real-world deployment and is largely confined to small-scale or permissioned settings. Moreover, QKD does not provide Sybil resistance or incentive mechanisms and is therefore unsuitable as a foundation for permissionless blockchain consensus.

D. Quantum Blockchain and Proof-of-Quantum-Work

To replace classical hash inversion with quantum sampling tasks, Ref. [11] proposes a PoW consensus mechanism based on coarse-grained boson sampling, where miners generate photonic samples whose statistical properties are verifiable but classically hard to spoof. The protocol demonstrates that quantum devices can be used to construct PoW puzzles with strong complexity-theoretic foundations, and provides a Nash equilibrium analysis that incentivizes honest quantum sampling. PoQ has been proposed as a quantum-native alternative to classical PoW, requiring miners to perform quantum computations that are infeasible to simulate classically. Ref. [12] demonstrates a practical PoQ instantiation using quantum sampling tasks, along with probabilistic validation methods tolerant to quantum noise.

However, this line of work assumes a network in which mining power is predominantly quantum and does not consider coexistence with classical PoW miners or gradual deployment scenarios. It replaces classical PoW entirely rather than integrating with existing consensus, and does not analyze fork dynamics or chain selection when quantum and classical blocks compete. As a result, they leave open the problem of incentive-compatible transition from classical to quantum mining.

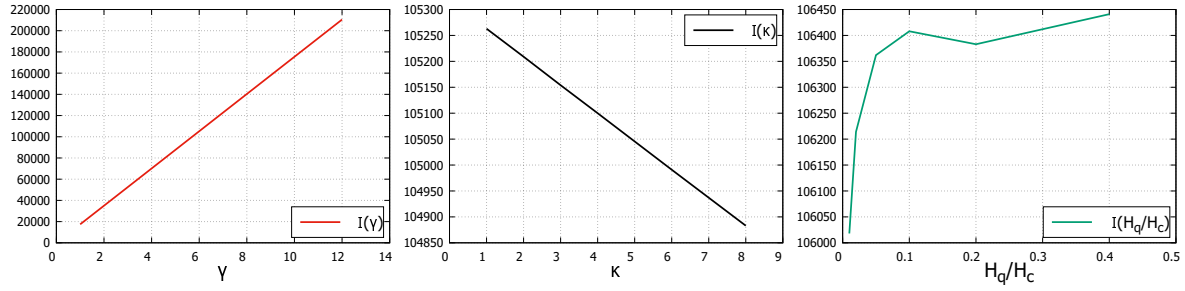


Fig. 4. The relationship between incentive received by PoQ miner versus other variables, (a) γ , (b) κ , (c) H_q/H_c .

VIII. CONCLUSION

This paper explores how quantum computation can be incorporated into blockchain consensus without disrupting existing systems. We presented a hybrid design that combines classical Proof-of-Work with periodic quantum-validated checkpoints, enabling quantum security benefits while preserving decentralization, verification efficiency, and deployment practicality. Our results show that hybrid consensus can stabilize validation cost, reduce fork instability compared to fully quantum approaches, and provide meaningful protection against asymmetric classical or quantum adversaries. More importantly, it offers a concrete migration path that does not require abandoning today's mining infrastructure or economic model. We view this design as a key step toward consensus protocols that evolve with computing technology rather than being tied to a single computational assumption. Future work will study large-scale deployment behavior, integration with post-quantum cryptographic primitives, and extensions to other consensus families.

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